

Forensic Examiner's Report

CYBR 472: Cybercrime Investigations — Capstone Investigation

Evidence Catalogue and the team's findings interpretation. Process documentation belongs in each examiner's individual Actions Taken log.

MEMO TO FILE

FORENSIC CASE NUMBER:	NZ-ECL-2025-0342
REQUESTING AGENCY:	New Zealand Customs Service, Drug Investigations Unit
OFFENCE:	Importation of Class A Controlled Drug (Methamphetamine)
LEGAL AUTHORITY:	Search Warrant SW-2025-1142; Customs and Excise Act 2018, Section 151
TEAM REFERENCE:	Chris & Sahil
RECEIVED:	20/04/2026
OPENED:	10/05/2026
COMPLETED:	22/05/2026
FORENSIC HOURS:	40
OS EXAMINED:	Windows 10 Pro
FILE SYSTEM:	NTFS
DATA ANALYSED:	EV-001 (desktop image, size 30 GB); EV-002 (memory dump files, size 16 GB),

Role Allocation

[Identify which team member is responsible for each specialist area. For teams of 1-2, indicate shared areas.]

Team Member	Specialist Area	Scope
Sahil Adwani	A: File Systems & Data Recovery	Primary
Chris Chivers	B: OS & Application Analysis	Primary
Sahil Adwani Chris Chivers	C: Communications & Web Activity	Shared

Tools and Software

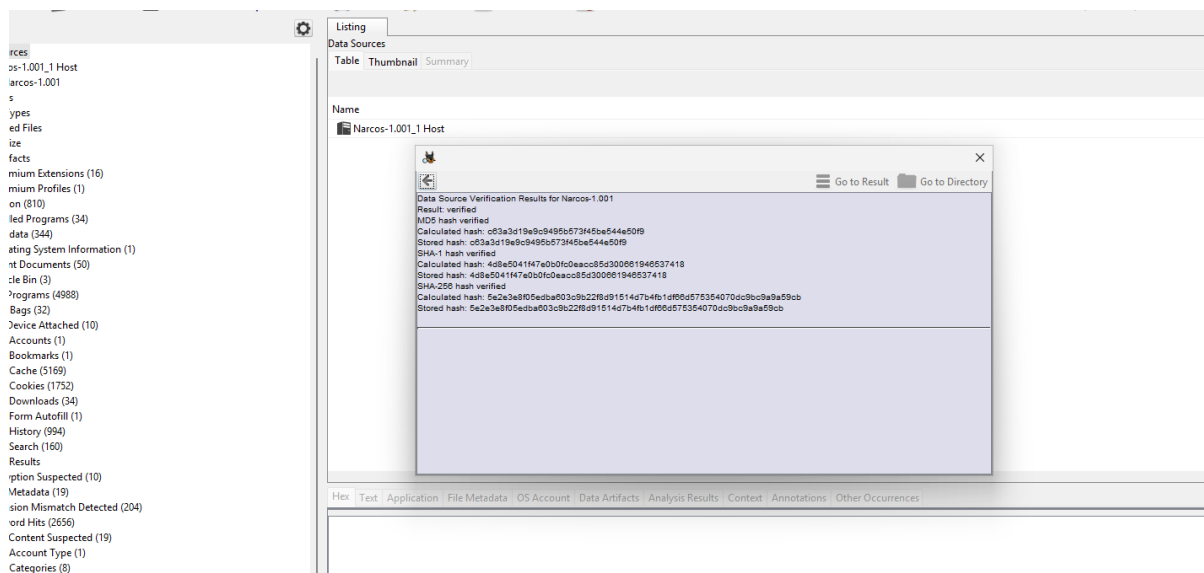
[List forensic tools used during the examination with version numbers.]

Tool	Version	Purpose
VirtualBox https://www.virtualbox.org/	7.2.6	Compartmentalization / Forensics Environment
Autopsy https://www.autopsy.com/	4.21.0	Primary forensic analysis suite

FTK Imager https://www.exterro.com/digital-forensics-software/ftk-imager	4.7.3.81	Forensic imaging / evidence verification
RegRipper https://github.com/keydet89/RegRipper3.0	3.0	Registry hive parsing
HxD https://mh-nexus.de/en/hxd/	2.5.0.0	Hex editor / file system analysis
RegistryExplorer https://www.sans.org/tools/registry-explorer	2.1.0	Search through registry hive files in a user-friendly way like file explorer
WinDbg https://learn.microsoft.com/en-us/windows-hardware/drivers/debugger/	1.2603.20001	Debugger for Windows applications, device drivers, and memory dumps
Windows CMD	10.0.26200.8457	Command shell interface for windows for executing various commands
Strings https://learn.microsoft.com/en-us/sysinternals/downloads/strings	2.54	Tool to output string contents of files on disk

Acquisition and verification of the disk image:

Data Source Verification Results for Narcos-1.001 using Autopsy:



Using FTK imager:

Drive/Image Verify Results	
[-]	
Name	Narcos-1.001
Sector count	62914560
[-] MD5 Hash	
Computed hash	c63a3d19e9c9495b573f45be544e50f9
Report Hash	c63a3d19e9c9495b573f45be544e50f9
Verify result	Match
[-] SHA1 Hash	
Computed hash	4d8e5041f47e0b0fc0eacc85d300661946537418
Report Hash	4d8e5041f47e0b0fc0eacc85d300661946537418
Verify result	Match
[-] Bad Blocks List	
Bad block(s) in image	No bad blocks found in image

Evidence Catalogue

[Under each examination task, list the evidence items the team has decided are significant, with file paths to the supporting archive. Describe what was found and where — do NOT interpret what the evidence means here. Interpretation belongs in the Findings Interpretation section. Include negative findings where relevant.]

Task 1: User Attribution

[List evidence relating to the owner/primary user of the desktop. Present the evidence neutrally.]

Registry Analysis

Regripper.exe shows details exposed by the windows SAM registry:

```
Username       : Steve [1001]
SID            : S-1-5-21-1474204758-2504895174-1356074821-1001
Full Name      :
User Comment    :
Account Type    :
Account Created : Mon Jan 28 19:35:06 2019 Z
Name           :
Last Login Date : Sat Feb  2 02:38:22 2019 Z
Pwd Reset Date  : Mon Jan 28 19:35:06 2019 Z
Pwd Fail Date   : Tue Jan 29 21:06:35 2019 Z
Login Count     : 18
--> Password does not expire
--> Password not required
--> Normal user account
```

The OS User Account 'Steve' has a count of 18 logins.

Command: rip.exe -r "SAM" -a sam > sam_analysis.txt

Location: /img_Narcos-1.001/vol_vol7/Windows/System32/config/SAM

Output: /Export/sam_analysis.txt

NTUSER.DAT

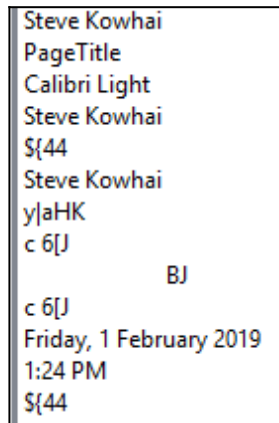
```
-----
UserAssist
Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist
LastWrite Time 2019-01-28 19:35:33Z

{9E04CAB2-CC14-11Df-BB8C-A2F1DED72085}
{A3D53349-6E61-4557-8FC7-0028EDCEE6F6}
{B267E3AD-A825-4A09-82B9-EEC22AA3B847}
{BC848336-4DD0-48FF-BB08-D3190DACB3E2}
{CAA59E3C-4792-41A5-9909-6A6A8D32490E}
{CEBFF5CD-ACE2-4F4F-9178-9926F41749EA}
2019-02-02 02:33:54Z
  com.squirrel.Discord.Discord (2)
2019-02-02 02:32:58Z
  Microsoft.Windows.Explorer (45)
2019-02-02 02:28:15Z
  Chrome (10)
2019-02-02 01:05:31Z
  Microsoft.ScreenSketch_8wekyb3d8bbwe!App (4)
2019-02-01 03:00:33Z
  E:\Project Digital Corpora\Memory Dump Tools\Comae\DumpIt.exe (2)
2019-02-01 02:56:13Z
  {6D809377-6AF0-444B-8957-A3773F02200E}\CCleaner\CCleaner64.exe (3)
2019-02-01 00:25:59Z
  Microsoft.Office.OneNote_8wekyb3d8bbwe!microsoft.onenoteim (2)
2019-02-01 00:16:46Z
  C:\Users\Steve\Downloads\Image Steganography 1.5.2 Setup.exe (1)
2019-01-31 03:08:52Z
  Microsoft.AutoGenerated.{8ABD94FB-E7D6-84A6-A997-C918EDDE0AE5} (1)
2019-01-31 03:08:35Z
  Microsoft.Windows.ControlPanel (1)
2019-01-29 23:41:03Z
  C:\Users\Steve\Downloads\ccsetup552.exe (1)
2019-01-29 23:31:23Z
  TrueCryptFoundation.TrueCrypt (1)
2019-01-29 23:29:17Z
  {7C5A40EF-A0FB-4BFC-874A-C0F2E0B9FA8E}\OpenOffice 4\program\swriter.exe (1)
2019-01-29 23:28:14Z
  C:\Users\Steve\Downloads\TrueCrypt Setup 7.1a.exe (1)
2019-01-29 21:05:15Z
  C:\Users\Steve\Downloads\ChromeSetup.exe (1)
2019-01-29 21:04:28Z
  Microsoft.MicrosoftEdge_8wekyb3d8bbwe!MicrosoftEdge (8)
2019-01-29 04:31:41Z
  E:\Memory Dump Tools\Comae\DumpIt.exe (1)
2019-01-29 01:06:15Z
  {7C5A40EF-A0FB-4BFC-874A-C0F2E0B9FA8E}\OpenOffice 4\program\soffice.exe (3)
2019-01-28 20:56:33Z
  C:\Users\Steve\Downloads\Misc\DiscordSetup.exe (1)
2019-01-28 20:43:48Z
```

Location: /img_Narcos-1.001/vol_vol7/Users/Steve/NTUSER.DAT

Output: /Export/SteveNTUSERreport.txt

00000006.bin



Location:

/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Local/Packages/Microsoft.Office.OneNote_8wekyb3d8bbwe/LocalState/AppData/Local/OneNote/16.0/cache/00000006.bin

Volatility Command Line Execution on Desktop

Command:

```
python C:\Users\Cyber\Desktop\DigitalForensicTools\volatility-2.6.1\vol.py  
-f DESKTOP-EN8JNOM-20190129-041409.dmp --profile=Win10x64_17134 cmdline
```

Filtered Output:

```
cscript.exe pid: 7352 Command line : cscript.exe "C:  
\Users\JohnF\AppData\Local\Temp\bmt7BB1.vbs" "C:  
\Users\JohnF\AppData\Local\Temp\VbsReprot.ini" "00000004" 2019-01-28  
20:18:21
```

```
Discord.exe pid: 8052 Command line : "C:  
\Users\JohnF\AppData\Local\Discord\app-0.0.304\Discord.exe" --  
squirrel-firstrun 2019-01-28 20:21:38
```

```
Discord.exe pid: 6676 Command line : C:  
\Users\JohnF\AppData\Local\Discord\app-0.0.304\Discord.exe --  
reporter-url=http://crash.discordapp.com:1127/post --application-  
name=Discord "--crashes-directory=C:  
\Users\JohnF\AppData\Local\Temp\Discord Crashes" --v=1 2019-01-28  
20:21:43
```

```
Discord.exe pid: 7128 Command line : C:  
\Users\JohnF\AppData\Local\Discord\app-0.0.304\Discord.exe --overlay-  
host 2019-01-28 20:21:56
```

vmacthlp.exe pid: 1596 Command line : "C:\Program Files\VMware\VMware Tools\vmacthlp.exe" 2019-01-28 19:32:37

vmtoolsd.exe pid: 2692 Command line : "C:\Program Files\VMware\VMware Tools\vmtoolsd.exe" 2019-01-28 19:32:39

VGAuthService.exe pid: 2760 Command line : "C:\Program Files\VMware\VMware Tools\VMware VGAuthService\VGAuthService.exe" 2019-01-28 19:32:39

vmtoolsd.exe pid: 6524 Command line : "C:\Program Files\VMware\VMware Tools\vmtoolsd.exe" -n vmusr 2019-01-28 19:33:19

BavSvc.exe pid: 7232 Command line : "C:\Program Files (x86)\Baidu Security\Baidu Antivirus\5.4.3.147185.0\BavSvc.exe" 2019-01-28 20:18:28

BHipsSvc.exe pid: 5456 Command line : "C:\Program Files (x86)\Baidu Security\Baidu Antivirus\5.4.3.147185.0\BHipsSvc.exe" 2019-01-28 20:18:30

bavhm.exe pid: 1860 Command line : "C:\Program Files (x86)\Baidu Security\Baidu Antivirus\5.4.3.147185.0\bavhm.exe" 2019-01-28 20:18:33

Extracted Images from Autopsy



Location: /img_Narcos-1.001/vol_vol7/Users/Steve/Pictures/620x349.jpg

Modified: 2019-02-01 15:48:41 NZDT

Changed: 2019-02-01 15:48:41 NZDT

Accessed: 2019-01-31 16:04:13 NZDT

Created: 2019-01-31 15:57:04 NZDT

\$RIIK1AS.jpg:

Location: /img_Narcos-1.001/vol_vol7/\$Recycle.Bin/S-1-5-21-1474204758-2504895174-1356074821-1001/\$RIIK1AS.jpg

Modified: 2019-01-31 15:57:06 NZDT

Changed: 2019-02-01 15:48:41 NZDT

Accessed: 2019-01-31 16:04:13 NZDT

Created: 2019-01-31 15:57:04 NZDT



Location: /img_Narcos-1.001/vol_vol7/Users/Steve/Pictures/price-meth-bust-4.jpg
price-meth-bust-4.jpg

Modified: 2019-02-01 15:48:41 NZDT

Changed: 2019-02-01 15:48:41 NZDT

Accessed: 2019-01-31 16:04:13 NZDT

Created: 2019-01-31 15:58:22 NZDT

\$RA3IE5E.jpg

Location: /img_Narcos-1.001/vol_vol7/\$Recycle.Bin/S-1-5-21-1474204758-2504895174-1356074821-1001/\$RA3IE5E.jpg

Modified: 2019-01-31 15:58:22 NZDT
Changed: 2019-02-01 15:48:41 NZDT
Accessed: 2019-01-31 16:04:13 NZDT
Created: 2019-01-31 15:58:22 NZDT



Location:
/img_Narcos-1.001/vol_vol7/Users/Steve/Pictures/eight_col_patches_crp.jpg

Modified: 2019-02-01 15:48:41 NZDT
Changed: 2019-02-01 15:48:41 NZDT
Accessed: 2019-02-01 15:42:46 NZDT
Created: 2019-01-31 15:59:38 NZDT

\$R5WIK39.jpg

Location: /img_Narcos-1.001/vol_vol7/\$Recycle.Bin/S-1-5-21-1474204758-2504895174-1356074821-1001/\$R5WIK39.jpg

Modified: 2019-01-31 15:59:38 NZDT
Changed: 2019-02-01 15:48:41 NZDT
Accessed: 2019-02-01 15:42:46 NZDT
Created: 2019-01-31 15:59:38 NZDT



Location: /img_Narcos-1.001/vol_vol7/\$CarvedFiles/1/f0332416.jpg



Location: /img_Narcos-1.001/vol_vol7/\$CarvedFiles/1/f0377496.jpg



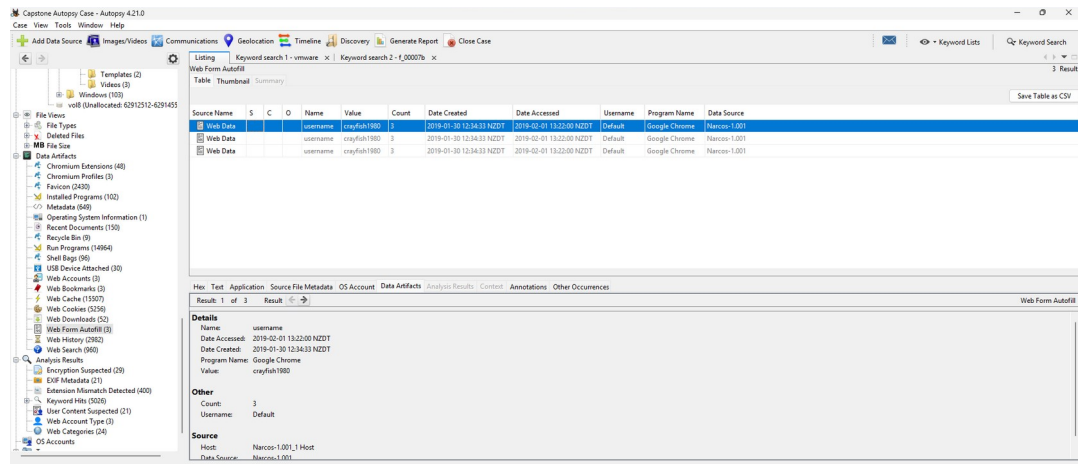
Location: /img_Narcos-1.001/vol_vol7/\$CarvedFiles/3/f0991144.jpg
(Pseudo Meth)

diving-crayfish-960x540.jpg



Location: /img_Narcos-1.001/vol_vol7/Users/Steve/Pictures/diving-crayfish-960x540.jpg

Location:
/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Roaming/Microsoft/Windows/Themes/TranscodedWallpaper



Location:

/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Local/Google/Chrome/
 User Data/Default/Web Data

Proton Mail username is displayed: “crayfish1980”

Task 2: Relationship with John Fredricksen

[List evidence relating to communications or connections between the desktop user and John Fredricksen.]

Discord Chat Logs

Location:

/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Roaming/Discord/Local
 Storage/leveldb/000006.txt

“New supplier eh? Definitely Interested! Can I get 10 keys of it delivered to

Wellington”

“Yeah yeah probably wiser, good one. In fact I have already put a document together in anticipation of chatting with you. I'll send it through now. It contains some information regarding how I see this working out. Let me know what you think once you have read it. S O”

“Good Thinking, I already know how. Heard of steganography?”

“A way of hiding one image within another. There's a simple application called 'Image Steganography’”

“Ya.. I just told you about the tool :face_palm: Received it. Will check to see if it works and confirm soon.”

“Meet at the Eastbourne library and if we miss each other come to 666 Rewera Avenue, Petone.”

Task 3: Drug Trafficking Network

[List evidence relating to other individuals or roles. Include negative findings where appropriate.]

Discord Chat Logs

The discord chat logs contain a “channel ID” identifying a specific “chat room”

Both the desktop user (Steve) and John’s laptop have Discord.exe processes that access the specific channel or (chat room). “539550615072800768” is the matching channel ID.

```
Owner: Process Discord.exe Pid 4700
0x4ealb6bd 35 33 39 35 35 30 36 31 35 30 37 32 38 30 30 37 5395506150728007
0x4ealb6cd 36 38 00 d1 66 c0 5b dd 41 70 4f dd 41 70 4f 0e 68..f.[.ApO.ApO.
0x4ealb6dd 00 00 00 59 41 70 4f 5d 20 36 48 a1 ea a1 4e 59 ...YApO].6H...NY
0x4ealb6ed 41 70 4f 59 41 70 4f 59 41 70 4f 00 00 00 00 59 ApOYApOYApO...Y
0x4ealb6fd 41 70 4f 49 ad a1 4e 49 ad a1 4e 59 41 70 4f 59 ApOI..NI..NYApOY
0x4ealb70d 41 70 4f 59 41 70 4f 00 00 00 00 10 00 00 00 59 ApOYApO.....Y
0x4ealb71d 41 70 4f 59 41 70 4f 59 41 70 4f 00 00 00 00 00 ApOYApOYApO....
0x4ealb72d 00 00 00 59 41 70 4f d1 66 c0 5b dd 41 70 4f dd ...YApO.f.[.ApO.
0x4ealb73d 41 70 4f 04 00 00 00 59 41 70 4f 79 b9 3b 3e ad ApO....YApOy.>.>.
0x4ealb74d ab a1 4e d9 ab a1 4e 3d ac a1 4e 59 41 70 4f 00 ..N...N=.NYApO.
0x4ealb75d 00 00 00 59 41 70 4f a1 ac a1 4e a1 ac a1 4e b5 ...YApO...N..N.
0x4ealb76d ac a1 4e 59 41 70 4f 59 41 70 4f 00 00 00 00 0a ..NYApOYApO....
0x4ealb77d 00 00 00 59 41 70 4f d1 b6 a1 4e 3d ac a1 4e 00 ...YApO...N=.N.
0x4ealb78d 00 00 00 00 00 00 e5 ac a1 4e 01 da 3c 57 dd .....N.<W.
0x4ealb79d 41 70 4f dd 41 70 4f b1 3d 0c 3e b5 ca 3e 3e f5 ApO.ApO.=.>.>.>.
0x4ealb7ad 03 e1 0f 4d d0 3e 3e fd 3d 0c 3e 89 3a 73 32 15 ...M.>>.>.>.:s2.
Rule: r1
```

Command: python C:

```
\Users\Cyber\Desktop\DigitalForensicTools\volatility-2.6.1\vol.py -f
DESKTOP-EN8JNOM-20190129-041409.dmp --profile=Win10x64_17134 yarascan
-Y "539550615072800768" (Steve's Desktop Memory)
```

```
Owner: Process Discord.exe Pid 684
0x5f3fca07 35 33 39 35 35 30 36 31 35 30 37 32 38 30 30 37 5395506150728007
0x5f3fca17 36 38 2f 74 79 70 69 6e 67 bd 4f b8 47 0c 00 00 68/typing.O.G...
0x5f3fca27 00 81 bd 3d 5f 99 bd 3d 5f 30 00 00 00 30 00 00 ...=...=...0...
0x5f3fca37 00 f1 d1 9e 35 61 cc 3f 5f c9 52 b8 47 03 00 00 ...5a.?_R.G...
0x5f3fca47 00 42 00 00 00 6f 00 77 00 6e 00 65 00 72 00 49 .B...o.w.n.e.r.I
0x5f3fca57 00 63 00 6f 00 6e 00 2d 00 31 00 63 00 41 00 52 .c.o.n.-.l.c.A.R
0x5f3fca67 00 4f 00 30 00 20 00 6f 00 77 00 6e 00 65 00 72 .O.0...o.w.n.e.r
0x5f3fca77 00 49 00 63 00 6f 00 6e 00 2d 00 75 00 5a 00 36 .I.c.o.n.-.u.Z.6
0x5f3fca87 00 6d 00 45 00 37 00 00 00 bd 4f b8 47 06 00 00 .m.E.7...O.G...
0x5f3fca97 00 51 cc 3f 5f 8d 43 b8 47 8d 43 b8 47 71 f1 79 .Q.?_C.G.C.Gq.y
0x5f3fcaa7 3d dd 41 b8 47 dd 41 b8 47 09 f6 cd 57 65 98 04 =.A.G.A.G...We..
0x5f3fcab7 2f 59 41 b8 47 59 41 b8 47 45 c9 3f 5f 59 41 b8 /Y.A.GY.A.GE.?_YA
0x5f3fcac7 47 bd 4f b8 47 06 00 00 00 41 cc 3f 5f 8d 43 b8 G.O.G....A.?_C.
0x5f3fcaed 47 8d 43 b8 47 bd 4f b8 47 06 00 00 00 31 cc 3f G.C.G.O.G...1?
0x5f3fcae7 5f 8d 43 b8 47 8d 43 b8 47 c9 52 b8 47 03 00 00 _C.G.C.G.R.G...
0x5f3fcac7 00 3a 00 00 00 6e 00 61 00 6d 00 65 00 54 00 61 .....n.a.m.e.T.a
Rule: r1
```

Command: python C:

```
\Users\Cyber\Desktop\DigitalForensicTools\volatility-2.6.1\vol.py -f
JOHNFLAPTOP1-20190130-020349.dmp --profile=Win10x64_17134 yarascan -Y
"539550615072800768" (John's Laptop Memory)
```

Using FTK Imager, we can find some discord log communication between John and Steve (The Desktop User)

Capstone Autopsy Case - Autopsy 4.21.0

Case View Tools Window Help

Listing Keyword search 1 - Firefox

Table Thumbnail

Page 1 of 1 Pages: Go to Page: Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Di)	Flags(Md)	Known	Location
[current folder]				2019-02-02 15:38:41 NZDT	2019-02-02 15:38:41 NZDT	2019-02-02 15:38:41 NZDT	2019-01-29 09:57:06 NZDT	96	Allocated	Allocated	unknown	/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Ro...
[parent folder]				2019-01-29 09:57:06 NZDT	2019-01-29 09:57:06 NZDT	2019-02-02 15:39:25 NZDT	2019-01-29 09:57:06 NZDT	144	Allocated	Allocated	unknown	/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Ro...
000006.txt			0	2019-01-29 13:12:06 NZDT	2019-01-29 13:12:06 NZDT	2019-02-02 15:39:25 NZDT	2019-01-29 13:12:06 NZDT	9533	Allocated	Allocated	unknown	/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Ro... cded1c1f3
000006.log			0	2019-02-02 15:38:47 NZDT	2019-02-02 15:38:47 NZDT	2019-02-02 15:38:47 NZDT	2019-01-29 16:43:06 NZDT	34411	Allocated	Allocated	unknown	/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Ro... 19da7c7c
000007.ab			0	2019-01-29 16:43:06 NZDT	2019-01-29 16:43:06 NZDT	2019-02-02 15:39:25 NZDT	2019-01-29 16:43:06 NZDT	7837	Allocated	Allocated	unknown	/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Ro... 59927eb35
CURRENT			1	2019-01-29 09:57:06 NZDT	2019-01-29 09:57:06 NZDT	2019-02-02 15:38:41 NZDT	2019-01-29 09:57:06 NZDT	16	Allocated	Allocated	unknown	/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Ro... 46295c4d
LOCK			0	2019-01-29 09:57:06 NZDT	2019-01-29 09:57:06 NZDT	2019-01-29 09:57:06 NZDT	2019-01-29 09:57:06 NZDT	0	Allocated	Allocated	unknown	/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Ro... 4d4dcd69
LOG			0	2019-02-02 15:38:41 NZDT	2019-02-02 15:38:41 NZDT	2019-02-02 15:39:25 NZDT	2019-01-29 09:57:06 NZDT	182	Allocated	Allocated	unknown	/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Ro... aadad62d
LOG.old			0	2019-02-02 13:39:54 NZDT	2019-02-02 15:38:41 NZDT	2019-02-02 15:38:41 NZDT	2019-01-29 09:57:06 NZDT	182	Allocated	Allocated	unknown	/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Ro... ccc78834
LOG.old-RF18bad.TMP				2019-01-30 09:59:44 NZDT	2019-01-31 09:41:41 NZDT	2019-01-30 10:02:48 NZDT	2019-01-29 09:57:06 NZDT	182	Unallocated	Unallocated	unknown	/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Ro... 888f8f8f

Has Test Application File Metadata OS Account Data Artifacts Analysis Results Content Annotations Other Occurrences

Strings: Extracted Text Translated

Page 1 of 1 Page Matches on page - of - Match 100% Reset Test Source: File Text

METAhttps://discordapp.com
https://discordapp.com
SelectedChannelName
["selectedVoiceChannelId":"539550615575986207","lastConnectedTime":"1548733352596","selectedChannelId":"539550614603168912","539550615072800768"]
METAhttps://discordapp.com
SelectedChannelName
https://discordapp.com
SelectedVoiceChannelId
["selectedVoiceChannelId":"539550615575986207","lastConnectedTime":"1548733412559","selectedChannelId":"539550614603168912","539550615072800768"]
METAhttps://discordapp.com
https://discordapp.com
SelectedChannelName
["selectedVoiceChannelId":"539550615575986207","lastConnectedTime":"1548733412559","selectedChannelId":"539550614603168912","539550615072800768"]
METAhttps://discordapp.com
https://discordapp.com
SelectedChannelName

Location:

/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Roaming/Discord/Local Storage/leveldb/000006.txt

Output: /Export/000006.txt

New supplier eh? Definitely Interested! Can I get 10 keys of it delivered to Wellington"

"Yeah yeah probably wiser, good one. In fact I have already put a document together in anticipation of chatting with you. I'll send it through now. It contains some information regarding how I see this working out. Let me know what you think once you have read it. S O"

"Good Thinking, I already know how. Heard of steganography?"

"A way of hiding one image within another. There's a simple application called 'Image Steganography"

"Ya.. I just told you about the tool :face_palm: Received it. Will check to see if it works and confirm soon."

"Meet at the Eastbourne library and if we miss each other come to 666 Rewera Avenue, Petone."

Discord Application Artifacts

Location: /img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Local/Discord

Location: C:/Users/JohnF/AppData/Local/Discord/app-0.0.304/Discord.exe (from Day 2 Laptop memory)

Command: python C:

\Users\Cyber\Desktop\DigitalForensicTools\volatility-2.6.1\vol.py -f JOHNFLAPTOP1-20190130-020349.dmp --profile=Win10x64_17134 cmdline | findstr /i "Discord"

Output:

```
Volatility Foundation Volatility Framework 2.6.1
Discord.exe pid: 8548
Command line : "C:\Users\JohnF\AppData\Local\Discord\app-0.0.304\Discord.exe"
Discord.exe pid: 8496
Command line : "C:\Users\JohnF\AppData\Local\Discord\app-0.0.304\Discord.exe" --type=gpu-process --enable-features=SharedArrayBuffer --no-sandbox --gpu-preferences=KAAAAAAAAACAAwBAAQAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAADAAAAAAAAAAAAAAAAAAAAAAQAAAAAAAAAAAAAAAAAAAAAAACwAAABAAAAAAAAAAAAQAAAAQAAAAQAAAAAAAAAAAAAALAAAA --service-request-channel-token=93714678870186266
49 --mojo-platform-channel-handle=1596 /prefetch:2
Discord.exe pid: 256
Command line : C:\Users\JohnF\AppData\Local\Discord\app-0.0.304\Discord.exe --reporter-url=http://crash.discordapp.com:1127/post --application-name=Discord --crashes-directory=C:\Users\JohnF\AppData\Loc
al\Temp\Discord Crashes" --v=1
Discord.exe pid: 604
Discord.exe pid: 4812
Discord.exe pid: 2252
Command line : C:\Users\JohnF\AppData\Local\Discord\app-0.0.304\Discord.exe --overlay-host
Discord.exe pid: 7888
Command line : "C:\Users\JohnF\AppData\Local\Discord\app-0.0.304\Discord.exe" --type=gpu-process --enable-features=SharedArrayBuffer --no-sandbox --gpu-preferences=KAAAAAAAAACAAwBAAQAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAADAAAAAAAAAAAAAAAAAAAAAAQAAAAAAAAAAAAAAAAAAAAAAACwAAABAAAAAAAAAAAAQAAAAQAAAAQAAAAAAAAAAAAAALAAAA --use-gl=swiftshader-webgl --service-request-chan
nel-token=6872101489388198574 --mojo-platform-channel-handle=1592 /prefetch:2
Discord.exe pid: 7216
Discord.exe pid: 3792
```

Volatility Process Tree Execution on Desktop

Command: python C:

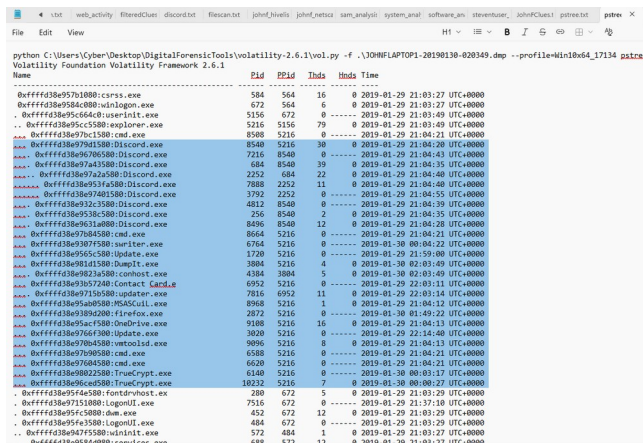
```
\Users\Cyber\Desktop\DigitalForensicTools\volatility-2.6.1\vol.py  
-f .\DESKTOP-EN8JNOM-20190129-041409.dmp --profile=Win10x64_17134  
pstree
```

Output:

```
DiscordSetup.exe 4692 6596 0 ----- 2019-01-28 20:21:15 UTC+0000
```

John's Laptop Day 2

Discord.exe, TrueCrypt.exe, VMtoolsd.exe running.



Name	Pid	PPid	Thds	Hnds	Time
0xfffff38e957b1080:csrss.exe	584	564	16	0	2019-01-29 21:03:27 UTC+0000
0xfffff38e95840800:winlogon.exe	572	564	6	0	2019-01-29 21:03:27 UTC+0000
0xfffff38e95664000:userinit.exe	5156	572	0	0	2019-01-29 21:03:49 UTC+0000
0xfffff38e95c55800:explorer.exe	5216	5156	79	0	2019-01-29 21:03:49 UTC+0000
0xfffff38e977c1380:cmd.exe	8508	5216	0	0	2019-01-29 21:04:21 UTC+0000
0xfffff38e97855800:Discord.exe	8540	5216	30	0	2019-01-29 21:04:28 UTC+0000
0xfffff38e97855800:Discord.exe	7216	8540	0	0	2019-01-29 21:04:43 UTC+0000
0xfffff38e97a41580:Discord.exe	684	8540	39	0	2019-01-29 21:04:35 UTC+0000
0xfffff38e97a2a580:Discord.exe	2252	684	22	0	2019-01-29 21:04:48 UTC+0000
0xfffff38e953fa580:Discord.exe	7888	2252	11	0	2019-01-29 21:04:48 UTC+0000
0xfffff38e9784b580:Discord.exe	3792	2252	0	0	2019-01-29 21:04:55 UTC+0000
0xfffff38e932c3580:Discord.exe	4812	8540	0	0	2019-01-29 21:04:39 UTC+0000
0xfffff38e953c5800:Discord.exe	256	8540	2	0	2019-01-29 21:04:35 UTC+0000
0xfffff38e9631a800:Discord.exe	8496	8540	12	0	2019-01-29 21:04:28 UTC+0000
0xfffff38e978b4580:cmd.exe	8664	5216	0	0	2019-01-29 21:04:21 UTC+0000
0xfffff38e9397f580:vmtoolsd.exe	6764	5216	0	0	2019-01-30 00:04:22 UTC+0000
0xfffff38e9565c580:update.exe	1720	5216	0	0	2019-01-29 21:59:00 UTC+0000
0xfffff38e981d1580:cmd.exe	3884	5216	4	0	2019-01-30 02:03:49 UTC+0000
0xfffff38e923a580:conhost.exe	4384	3884	5	0	2019-01-30 02:03:49 UTC+0000
0xfffff38e9357240:Contact Card.exe	6952	5216	0	0	2019-01-29 22:03:11 UTC+0000
0xfffff38e9715b580:update.exe	7816	6952	11	0	2019-01-29 22:03:16 UTC+0000
0xfffff38e95a8580:MSACSL.exe	8968	5216	1	0	2019-01-29 21:04:12 UTC+0000
0xfffff38e9389d200:firefox.exe	2872	5216	0	0	2019-01-30 01:49:22 UTC+0000
0xfffff38e95c4f580:cmd.exe	9108	5216	16	0	2019-01-29 21:04:13 UTC+0000
0xfffff38e976d7300:update.exe	3020	5216	0	0	2019-01-29 22:14:40 UTC+0000
0xfffff38e978b4580:vmtoolsd.exe	9696	5216	8	0	2019-01-29 21:04:13 UTC+0000
0xfffff38e9785580:cmd.exe	6280	5216	0	0	2019-01-29 21:04:21 UTC+0000
0xfffff38e978b4580:cmd.exe	6620	5216	0	0	2019-01-29 21:04:21 UTC+0000
0xfffff38e98021580:TrueCrypt.exe	6140	5216	0	0	2019-01-30 00:03:17 UTC+0000
0xfffff38e96cc6580:TrueCrypt.exe	10232	5216	7	0	2019-01-30 00:00:27 UTC+0000
0xfffff38e9574e580:fontdrvhost.exe	280	672	5	0	2019-01-29 21:03:29 UTC+0000
0xfffff38e9715b80:LogonUI.exe	7316	672	0	0	2019-01-29 21:37:18 UTC+0000
0xfffff38e95c5800:dm.exe	452	672	12	0	2019-01-29 21:03:29 UTC+0000
0xfffff38e95c5800:LogonUI.exe	484	672	0	0	2019-01-29 21:03:29 UTC+0000
0xfffff38e97f5580:userinit.exe	572	484	1	0	2019-01-29 21:03:27 UTC+0000
0xfffff38e95840800:services.exe	688	572	12	0	2019-01-29 21:03:27 UTC+0000

John's Laptop Day 3

Discord.exe, VMtoolsd.exe running.

Name	Pid	PPid	Thds	Hnds	Time
0xfffff888377b2c40:csrss.exe	596	576	16	0	2019-01-30 07:40:36 UTC+0000
0xfffff88836e40800:winlogon.exe	684	576	4	0	2019-01-30 07:40:36 UTC+0000
0xfffff88837929800:dm.exe	512	684	10	0	2019-01-30 07:40:38 UTC+0000
0xfffff88837929800:LogonUI.exe	532	684	0	0	2019-01-30 07:40:38 UTC+0000
0xfffff88834805800:userinit.exe	5252	684	0	0	2019-01-30 07:42:07 UTC+0000
0xfffff88834805800:explorer.exe	5288	5252	68	0	2019-01-30 07:42:07 UTC+0000
0xfffff88836c1a800:MSACSL.exe	1196	5288	1	0	2019-01-30 07:42:40 UTC+0000
0xfffff88836b48580:vmtoolsd.exe	7836	5288	0	0	2019-01-31 02:24:14 UTC+0000
0xfffff888351b0580:cmd.exe	5024	5288	0	0	2019-01-31 02:24:14 UTC+0000
0xfffff88838f4a580:conhost.exe	2784	5924	5	0	2019-01-31 02:24:14 UTC+0000
0xfffff88836a0d000:cmd.exe	7136	5288	16	0	2019-01-30 07:42:41 UTC+0000
0xfffff88838ac4580:firefox.exe	9452	5288	0	0	2019-01-31 01:50:18 UTC+0000
0xfffff88834855800:Discord.exe	8692	5288	38	0	2019-01-30 07:42:46 UTC+0000
0xfffff888379c4580:Discord.exe	5248	8692	1	0	2019-01-30 07:42:46 UTC+0000
0xfffff88835f6a800:Discord.exe	9864	8692	2	0	2019-01-30 07:43:01 UTC+0000
0xfffff888379d4800:Discord.exe	8996	8692	0	0	2019-01-30 07:43:01 UTC+0000
0xfffff88838880580:Discord.exe	7796	8996	22	0	2019-01-30 07:43:06 UTC+0000
0xfffff88838b22480:Discord.exe	8476	7796	11	0	2019-01-30 07:43:07 UTC+0000
0xfffff888379a5800:Discord.exe	9156	7796	0	0	2019-01-30 07:43:12 UTC+0000
0xfffff88835a05800:Discord.exe	18148	8692	0	0	2019-01-30 07:43:03 UTC+0000
0xfffff88835a05800:Discord.exe	5116	8692	0	0	2019-01-30 07:43:05 UTC+0000
0xfffff88835fa8580:cmd.exe	6116	5288	0	0	2019-01-30 07:42:13 UTC+0000
0xfffff88836b0c580:update.exe	8788	5288	10	0	2019-01-30 07:42:47 UTC+0000
0xfffff888377a5800:fontdrvhost.exe	980	684	5	0	2019-01-30 07:40:37 UTC+0000
0xfffff88835c34400:System	4	0	132	0	2019-01-30 07:40:14 UTC+0000
0xfffff88835453400:Registry	184	4	3	0	2019-01-30 07:40:02 UTC+0000
0xfffff88837850400:MemCompression	1824	4	46	0	2019-01-30 07:40:40 UTC+0000
0xfffff88836b0a040:sss.exe	308	4	2	0	2019-01-30 07:40:14 UTC+0000
0xfffff888367e4080:vmtoolsd.exe	584	496	1	0	2019-01-30 07:40:36 UTC+0000
0xfffff888361f0800:services.exe	724	584	12	0	2019-01-30 07:40:36 UTC+0000
0xfffff888380c3c00:svchost.exe	3888	724	3	0	2019-01-30 07:40:44 UTC+0000
0xfffff88837d25480:svchost.exe	2864	724	13	0	2019-01-30 07:40:41 UTC+0000
0xfffff88837f55580:audiodg.exe	5416	2864	5	0	2019-01-31 02:23:56 UTC+0000
0xfffff88837ac4580:svchost.exe	1556	724	2	0	2019-01-30 07:40:39 UTC+0000
0xfffff88836c3d580:svchost.exe	5656	724	5	0	2019-01-30 07:42:10 UTC+0000
0xfffff88837962580:svchost.exe	1828	724	4	0	2019-01-30 07:40:38 UTC+0000
0xfffff88838a73580:SignalBroker.exe	6688	724	2	0	2019-01-30 07:43:06 UTC+0000
0xfffff88835230580:wdmHost.exe	1748	724	0	0	2019-01-31 02:23:56 UTC+0000
0xfffff88837a05800:svchost.exe	1576	724	7	0	2019-01-30 07:40:39 UTC+0000
0xfffff888340a1580:taskhostw.exe	1668	1576	9	0	2019-01-30 07:42:05 UTC+0000
0xfffff88837905800:svchost.exe	1872	724	0	0	2019-01-30 07:40:38 UTC+0000
0xfffff888378f6480:svchost.exe	1092	724	4	0	2019-01-30 07:40:38 UTC+0000
0xfffff888377a2580:svchost.exe	88	724	6	0	2019-01-30 07:40:38 UTC+0000
0xfffff88837495800:svchost.exe	2744	724	16	0	2019-01-30 07:40:43 UTC+0000
0xfffff88837905800:svchost.exe	1624	724	0	0	2019-01-30 07:40:39 UTC+0000
0xfffff88834475580:SearchIndexer.exe	8288	724	21	0	2019-01-30 07:42:38 UTC+0000
0xfffff88837ae6580:svchost.exe	2688	724	5	0	2019-01-30 07:40:43 UTC+0000
0xfffff888377a2580:svchost.exe	876	724	28	0	2019-01-30 07:40:37 UTC+0000
0xfffff888341d6800:SkyperBackground	744	876	4	0	2019-01-30 08:02:19 UTC+0000
0xfffff888383dd000:smartscreen.exe	4208	876	11	0	2019-01-31 02:24:12 UTC+0000
0xfffff888387dc080:MicrosoftEdgeC	8832	876	14	0	2019-01-30 07:42:29 UTC+0000
0xfffff88836c14800:RuntimeBroker.exe	9592	876	2	0	2019-01-30 07:44:29 UTC+0000
0xfffff88836a0d000:cmdHost.exe	7708	876	5	0	2019-01-30 07:42:29 UTC+0000
0xfffff88837b05800:MicrosoftEdgeC	8848	876	16	0	2019-01-30 07:42:29 UTC+0000
0xfffff88836c6f080:ApplicationFra	1676	876	7	0	2019-01-30 07:42:27 UTC+0000

John's Laptop Day 4

Discord.exe and Image Steganography running.

Name	Ptd	PPtd	Inhs	Hnds	Time
0xffffb88377bc248:csrss.exe	596	576	16	0	2019-01-30 07:40:36 UTC+0000
0xffffb88364a6080:winlogon.exe	684	576	5	0	2019-01-30 07:40:36 UTC+0000
0xffffb8837929080:dw.exe	512	684	10	0	2019-01-30 07:40:38 UTC+0000
0xffffb883792d080:LogonUI.exe	532	684	0	-----	2019-01-30 07:40:38 UTC+0000
0xffffb883486b580:userinit.exe	5252	584	0	-----	2019-01-30 07:42:07 UTC+0000
0xffffb8834869580:explorer.exe	5288	5252	72	0	2019-01-30 07:42:07 UTC+0000
0xffffb8836c1a080:MSAScul.exe	1196	5288	1	0	2019-01-30 07:42:40 UTC+0000
0xffffb8839522580:Update.exe	9844	5288	0	-----	2019-01-31 21:02:01 UTC+0000
0xffffb8836b48580:vmtoolsd.exe	7036	5288	7	0	2019-01-30 07:42:40 UTC+0000
0xffffb883a2c1580:firefox.exe	8056	5288	0	-----	2019-02-01 00:04:01 UTC+0000
0xffffb88346b580:OneDrive.exe	7136	5288	16	0	2019-01-30 07:42:41 UTC+0000
0xffffb88351c2080:firefox.exe	3212	5288	0	-----	2019-01-31 21:25:20 UTC+0000
0xffffb8838ac4580:firefox.exe	9452	5288	0	-----	2019-01-31 01:59:18 UTC+0000
0xffffb8834639580:Discord.exe	8692	5288	30	0	2019-01-30 07:42:46 UTC+0000
0xffffb88375c4580:Discord.exe	9208	8692	12	0	2019-01-30 07:42:48 UTC+0000
0xffffb8835dfe080:Discord.exe	9964	8692	2	0	2019-01-30 07:43:01 UTC+0000
0xffffb88375b4800:Discord.exe	9996	8692	30	0	2019-01-30 07:43:01 UTC+0000
0xffffb8833080980:Discord.exe	7706	9096	22	0	2019-01-30 07:43:06 UTC+0000
0xffffb88338b2d080:Discord.exe	8476	7796	11	0	2019-01-30 07:43:07 UTC+0000
0xffffb8836aa7580:Discord.exe	9156	7796	0	-----	2019-01-30 07:43:22 UTC+0000
0xffffb8833aa9580:Discord.exe	10140	8692	0	-----	2019-01-30 07:43:03 UTC+0000
0xffffb8833ab580:Discord.exe	8128	8692	0	-----	2019-01-30 07:43:05 UTC+0000
0xffffb883522580:firefox.exe	3916	5288	0	-----	2019-01-31 22:59:06 UTC+0000
0xffffb883f9d580:DumpIt.exe	2516	5288	4	0	2019-02-01 02:04:01 UTC+0000
0xffffb883a49f580:conhost.exe	4060	2516	5	0	2019-02-01 02:04:01 UTC+0000
0xffffb88391e9580:firefox.exe	3544	5288	0	-----	2019-01-31 23:59:26 UTC+0000
0xffffb883952a580:ImageSteganog	8472	5288	0	-----	2019-01-31 23:00:43 UTC+0000
0xffffb8835fa580:led4unit.exe	6116	5288	0	-----	2019-01-30 07:42:13 UTC+0000
0xffffb8836b0c580:svchost.exe	8788	5288	12	0	2019-01-30 07:42:47 UTC+0000
0xffffb8837a0580:fontdrvhost.exe	908	684	5	0	2019-01-30 07:40:37 UTC+0000
0xffffb8833cb3440:System	4	0	126	0	2019-01-30 07:40:14 UTC+0000
0xffffb8833d53040:Registry	104	4	3	0	2019-01-30 07:40:02 UTC+0000
0xffffb8837b85040:MemCompression	1824	4	46	0	2019-01-30 07:40:40 UTC+0000
0xffffb8836b0b040:smss.exe	388	4	2	0	2019-01-30 07:40:14 UTC+0000
0xffffb8833ec5d00:svchost.exe	584	496	1	0	2019-01-30 07:40:36 UTC+0000
0xffffb88361fc000:services.exe	724	584	12	0	2019-01-30 07:40:36 UTC+0000
0xffffb883802c1c0:svchost.exe	3080	724	3	0	2019-01-30 07:40:44 UTC+0000
0xffffb8837d25400:svchost.exe	2064	724	13	0	2019-01-30 07:40:41 UTC+0000
0xffffb8839f70000:audiodg.exe	4964	2064	5	0	2019-02-01 02:03:30 UTC+0000
0xffffb8837ac4580:svchost.exe	1556	724	2	0	2019-01-30 07:40:39 UTC+0000
0xffffb883580c000:svchost.exe	5656	724	6	0	2019-01-30 07:42:10 UTC+0000
0xffffb8837902580:svchost.exe	1028	724	4	0	2019-01-30 07:40:38 UTC+0000
0xffffb8838a73580:SgrmBroker.exe	6688	724	2	0	2019-01-30 07:43:06 UTC+0000
0xffffb8837ad3580:svchost.exe	1576	724	11	0	2019-01-30 07:40:39 UTC+0000
0xffffb88340a1580:taskhostw.exe	1668	1576	8	0	2019-01-30 07:42:05 UTC+0000
0xffffb8837900580:svchost.exe	1072	724	8	0	2019-01-30 07:40:38 UTC+0000
0xffffb8838ea080:svchost.exe	172	724	7	0	2019-01-31 21:01:53 UTC+0000

Web Search

2019-01-28 20:34 – Viewed MetService (Wellington / Lower Hutt weather)
2019-01-28 20:55 – Searched “Discord” and downloaded DiscordSetup.exe
2019-01-28 22:59 – Searched “cutting agent” on Wikipedia
2019-01-28 23:00 – Searched “drug cutting”
2019-01-28 23:00 – Searched “cause effects”
2019-01-28 23:00 – Searched “cutting agent drug manufacturing”
2019-01-28 23:00 – Searched “what is meth cut with”
2019-01-29 01:04 – Searched “international drug routes”
2019-01-29 01:03 – Searched drug routes around Wellington
2019-01-31 15:55 – Searched “drug facts on crystal meth”
2019-02-02 14:01 – Searched “best places to trade drugs”

Proton Mail Commination

Access to Proton Mail

The screenshot shows the Autopsy 4.21.0 interface. The 'Web Accounts' section is expanded, showing search results for a Proton Mail account. The results table includes columns for Source Name, S, C, O, URL, Data Created, Decoded URL, Username, Realm, Domain, Program Name, and Data Source. The results show a Proton Mail login attempt on 2019-02-01 13:12:51 NZDT, with the URL https://mail.protonmail.com/login, Username protonmail.com, and Password protonmail.com.

Source Name	S	C	O	URL	Data Created	Decoded URL	Username	Realm	Domain	Program Name	Data Source	
Proton Mail				https://mail.protonmail.com/login	2019-02-01 13:12:51 NZDT	https://mail.protonmail.com/login	protonmail.com	Default	https://mail.protonmail.com/	protonmail.com	Google Chrome	Narcos-1.001
Login Data				https://mail.protonmail.com/login	2019-02-01 13:12:51 NZDT	protonmail.com	Default	https://mail.protonmail.com/	protonmail.com	Google Chrome	Narcos-1.001	

The 'Web Accounts' section is expanded, showing details for a Proton Mail account. The results table includes columns for Type, URL, Value, and Source(s). The results show a Proton Mail login attempt on 2019-02-01 13:12:51 NZDT, with the URL https://mail.protonmail.com/login, Username protonmail.com, and Password protonmail.com.

Type	URL	Value	Source(s)
URL	https://mail.protonmail.com/login		Recent Activity
Date Created	2019-02-01 13:12:51 NZDT		Recent Activity
Decoded URL	protonmail.com		Recent Activity
Username	https://mail.protonmail.com/		Recent Activity
Realm	protonmail.com		Recent Activity
Domain	protonmail.com		Recent Activity
Program Name	Google Chrome		Recent Activity
Source File Path	/img/Narcos-1.001/vol7/Users/Steve/AppData/Local/Google/Chrome/User Data/Default/Login Data		Recent Activity
Artifact ID	-8023370880474014		

Location:

/img_Narcos-1.001/vol7/Users/Steve/AppData/Local/Google/Chrome/
User Data/Default/Login Data

Proton Mail evidence for Steve.

```
Owner: Process svchost.exe Pid 3572
0x1b86abe5133 70 72 6f 74 6f 6e 6d 61 69 6c 2e 63 6f 6d 2f 69 protonmail.com/i
0x1b86abe5143 6e 62 6f 78 06 00 00 00 00 00 00 00 00 00 00 00 nbox.....
0x1b86abe5153 00 00 00 00 00 5c 4f 70 79 5c 76 fd 79 7b 22 74 .....\\Opy\\v.y{"t
0x1b86abe5163 79 70 65 22 3a 22 55 73 65 72 45 6e 67 61 67 65 type":"UserEngage
0x1b86abe5173 64 22 2c 22 72 65 70 6f 72 74 69 6e 67 41 70 70 d","reportingApp
0x1b86abe5183 22 3a 22 53 68 65 6c 6c 41 63 74 69 76 69 74 79 ": "ShellActivity
0x1b86abe5193 4d 6f 6e 69 74 6f 72 22 2c 22 61 63 74 69 76 65 Monitor","active
0x1b86abe51a3 44 75 72 61 74 69 6f 6e 53 65 63 6f 6e 64 73 22 DurationSeconds"
0x1b86abe51b3 3a 31 2c 22 73 68 65 6c 6c 43 6f 6e 74 65 6e 74 :1,"shellContent
0x1b86abe51c3 44 65 73 63 72 69 70 74 69 6f 6e 22 3a 7b 22 4d Description":{"M
0x1b86abe51d3 65 72 67 65 64 47 61 70 22 3a 36 30 30 7d 2c 22 mergedGap":600},"
0x1b86abe51e3 75 73 65 72 54 69 6d 65 7a 6f 6e 65 22 3a 22 41 userTimezone":"A
0x1b86abe51f3 75 73 74 72 61 6c 69 61 2f 42 72 69 73 62 61 6e ustralia/Brisban
0x1b86abe5203 65 22 7d 45 4f 58 79 6c 48 37 51 51 32 41 53 4b e"}EOXylH7QQ2ASK
0x1b86abe5213 72 6c 32 43 49 4a 6c 6c 50 6d 45 33 33 4f 6f 47 r12Cij1lPmE330oG
0x1b86abe5223 62 72 47 51 49 55 38 57 78 31 79 43 68 41 3d 5c brGQIU8Wx1yChA=\\
Rule: r1
Owner: Process svchost.exe Pid 3572
0x1b86abe7a43 70 72 6f 74 6f 6e 6d 61 69 6c 2e 63 6f 6d 2f 69 protonmail.com/i
0x1b86abe7a53 6e 62 6f 78 05 00 00 00 00 00 00 00 00 00 00 00 nbox.....
0x1b86abe7a63 00 00 00 00 00 5c 4f 70 79 5c 76 fd 79 7b 22 64 .....\\Opy\\v.y{"d
0x1b86abe7a73 69 73 70 6c 61 79 54 65 78 74 22 3a 22 49 6e 62 isplayText":"Inb
0x1b86abe7a83 6f 78 20 7c 20 68 65 72 65 73 6a 6f 68 6e 6e 79 ox.|.heresjohnny
0x1b86abe7a93 31 40 70 72 6f 74 6f 6e 6d 61 69 6c 2e 63 6f 6d 1@protonmail.com
0x1b86abe7aa3 20 7c 20 50 72 6f 74 6f 6e 4d 61 69 6c 22 2c 22 .|.ProtonMail","
0x1b86abe7ab3 61 63 74 69 76 61 74 69 6f 6e 55 72 69 22 3a 22 activationUri":"
0x1b86abe7ac3 6d 69 63 72 6f 73 6f 66 74 2d 65 64 67 65 3a 68 microsoft-edge:h
0x1b86abe7ad3 74 74 70 73 3a 2f 2f 6d 61 69 6c 2e 70 72 6f 74 ttps://mail.prot
0x1b86abe7ae3 6f 6e 6d 61 69 6c 2e 63 6f 6d 2f 69 6e 62 6f 78 onmail.com/inbox
0x1b86abe7af3 22 2c 22 61 70 70 44 69 73 70 6c 61 79 4e 61 6d ", "appDisplayNam
0x1b86abe7b03 65 22 3a 22 4d 69 63 72 6f 73 6f 66 74 20 45 64 e":"Microsoft.Ed
0x1b86abe7b13 67 65 22 2c 22 64 65 73 63 72 69 70 74 69 6f 6e ge","description
0x1b86abe7b23 22 3a 22 68 74 74 70 73 3a 2f 2f 6d 61 69 6c 2e ": "https://mail.
0x1b86abe7b33 70 72 6f 74 6f 6e 6d 61 69 6c 2e 63 6f 6d 2f 69 protonmail.com/i
Rule: r1
Owner: Process svchost.exe Pid 3572
0x1b86abe7a95 70 72 6f 74 6f 6e 6d 61 69 6c 2e 63 6f 6d 20 7c protonmail.com.|
0x1b86abe7aa5 20 50 72 6f 74 6f 6e 4d 61 69 6c 22 2c 22 61 63 .ProtonMail","ac
0x1b86abe7ab5 74 69 76 61 74 69 6f 6e 55 72 69 22 3a 22 6d 69 tivationUri":"mi
0x1b86abe7ac5 63 72 6f 73 6f 66 74 2d 65 64 67 65 3a 68 74 74 crosoft-edge:htt
0x1b86abe7ad5 70 73 3a 2f 2f 6d 61 69 6c 2e 70 72 6f 74 6f 6e ps://mail.proton
0x1b86abe7ae5 6d 61 69 6c 2e 63 6f 6d 2f 69 6e 62 6f 78 22 2c mail.com/inbox",
0x1b86abe7af5 22 61 70 70 44 69 73 70 6c 61 79 4e 61 6d 65 22 "appDisplayName"
0x1b86abe7b05 3a 22 4d 69 63 72 6f 73 6f 66 74 20 45 64 67 65 : "Microsoft.Edge
0x1b86abe7b15 22 2c 22 64 65 73 63 72 69 70 74 69 6f 6e 22 3a ", "description":
```

A string search in the desktop memory using volatility and filtering for "protonmail", we can see that a user by the name of heresjohnny1 also has a Proton Mail account.

Location: C:\Users\Cyber\Downloads\John Primary\Day 1\DESKTOP-EN8JNOM-20190129-041409.dmp

Command: python C:\Users\Cyber\Desktop\DigitalForensicTools\volatility-2.6.1\vol.py -f DESKTOP-EN8JNOM-20190129-041409.dmp --profile=Win10x64_17134 yarascan -Y "protonmail" > ProtonMailClues.txt

Task 4: Customers

[List evidence potentially relating to customers of the trafficking network.]

Potential Customers



Location: /img_Narcos-1.001/vol_vol7/\$CarvedFiles/1/f0667000.jpg



Location:
/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Local/Google/Chrome/User
Data/Default/Cache/f_000948



Location:
/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Local/Google/Chrome/User
Data/Default/Cache/f_00094a

```

Owner: Process RuntimeBroker. Pid 7456
0x212d925fd3e 63 6c 69 65 6e 74 73 2e 6f 64 73 00 48 00 09 00 clients.ods.H...
0x212d925fd4e 04 00 ef be 3c 4e 45 a3 3c 4e 2f ae 2e 00 00 00 ....<NE.<N/.....
0x212d925fd5e 07 8f 01 00 00 00 02 00 00 00 00 00 00 00 00 .....
0x212d925fd6e 00 00 00 00 00 00 e5 a2 18 01 63 00 6c 00 69 00 .....c.l.i.
0x212d925fd7e 65 00 6e 00 74 00 73 00 2e 00 6f 00 64 00 73 00 e.n.t.s...o.d.s.
0x212d925fd8e 00 00 1a 00 46 00 00 00 1a 00 ef be 02 00 6f 00 ....F.....o.
0x212d925fd9e 70 00 65 00 6e 00 64 00 6f 00 63 00 75 00 6d 00 p.e.n.d.o.c.u.m.
0x212d925fdae 65 00 6e 00 74 00 2e 00 43 00 61 00 6c 00 63 00 e.n.t...C.a.l.c.
0x212d925fdb6 44 00 6f 00 63 00 75 00 6d 00 65 00 6e 00 74 00 D.o.c.u.m.e.n.t.
0x212d925fdce 2e 00 31 00 00 00 1a 00 1a 00 00 00 00 00 00 00 ..1.....
0x212d925fdd6 06 00 5c 00 52 00 45 00 47 00 49 00 53 00 54 00 ..\R.E.G.I.S.T.
0x212d925fdee 52 00 59 00 5c 00 55 00 53 00 45 00 52 00 5c 00 R.Y.\U.S.E.R.\.
0x212d925fdfe 53 00 2d 00 31 00 2d 00 35 00 2d 00 32 00 31 00 S.-.1.-.5.-.2.1.
0x212d925fe0e 2d 00 31 00 32 00 38 00 38 00 38 00 34 00 30 00 -.1.2.8.8.4.0.
0x212d925fe1e 39 00 34 00 34 00 2d 00 34 00 32 00 30 00 39 00 9.4.4.-.4.2.0.9.
0x212d925fe2e 33 00 38 00 30 00 32 00 32 00 37 00 2d 00 32 00 3.8.0.2.2.7.-.2.

```

Command used: python C:

```

\Users\Cyber\Desktop\DigitalForensicTools\volatility-2.6.1\vol.py -f
JOHNFLAPTOP1-20190130-020349.dmp --profile=Win10x64_17134 cmdline | findstr
/i "client.ods"

```

A file called client.ods exists in the memory of Steve's desktop and John's laptop.

Task 5: Timeline and Future Plans

[List evidence that establishes dates, times, and planned activities.]

Extracted Images from Autopsy

✓ Nice Job! You picked one of our cheapest flights.
Book now so you don't miss out on this price!

16 Feb. 2019 From **Brisbane, QLD (BNE) (BNE)**
To **Wellington Intl. (WLG)**

Virgin Australia

8:45 am → 3:15 pm 3h 30m, Direct
BNE WLG

Show flight and baggage fee details ✕

23 Feb. 2019 From **Wellington Intl. (WLG)**
To **Brisbane, QLD (BNE) (BNE)**

Qantas Airways

6:15 am → 5:40 pm 14h 25m, 1 stop
WLG BNE AKL

Show flight and baggage fee details ✕

Trip Summary

Traveller 1: Adult	AUS\$663.91
Flight	AUS\$470.00
Taxes & Fees	AUS\$193.91
Traveller 2: Adult	AUS\$663.91
Flight	AUS\$470.00
Taxes & Fees	AUS\$193.91
Booking Fee	AUS\$0.00

Trip Total From: **AUS\$1,327.82**
Only 7 tickets left at this price!

Rates are quoted in Australian dollars

Important Flight Information

- Your flight is a combination of two one-way fares, each subject to its own rules and restrictions. If one of your flights is changed or cancelled, it will not automatically alter the other flight. Changes to the other flight may incur a charge.

Departure

- Tickets are non-refundable and non transferable. Name changes are not allowed.
- There may be an additional fee based on your payment.

2 Plane tickets for 2 travellers: Brisbane to Wellington and Wellington to Brisbane.

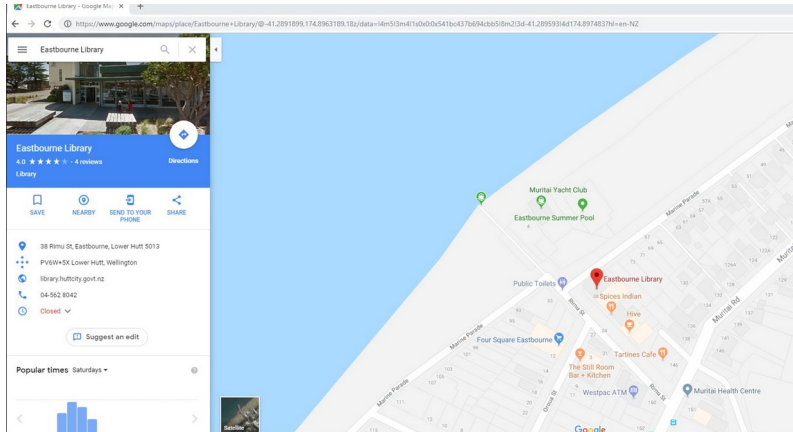
Location:

/img_Narcos-1.001/vol_vol7/Users/Steve/Documents/Misc/flightbookings.PNG

Modified: 2019-02-02 15:28:45 NZDT

Accessed: 2019-02-02 15:31:06 NZDT

Created: 2019-02-02 15:28:44 NZDT



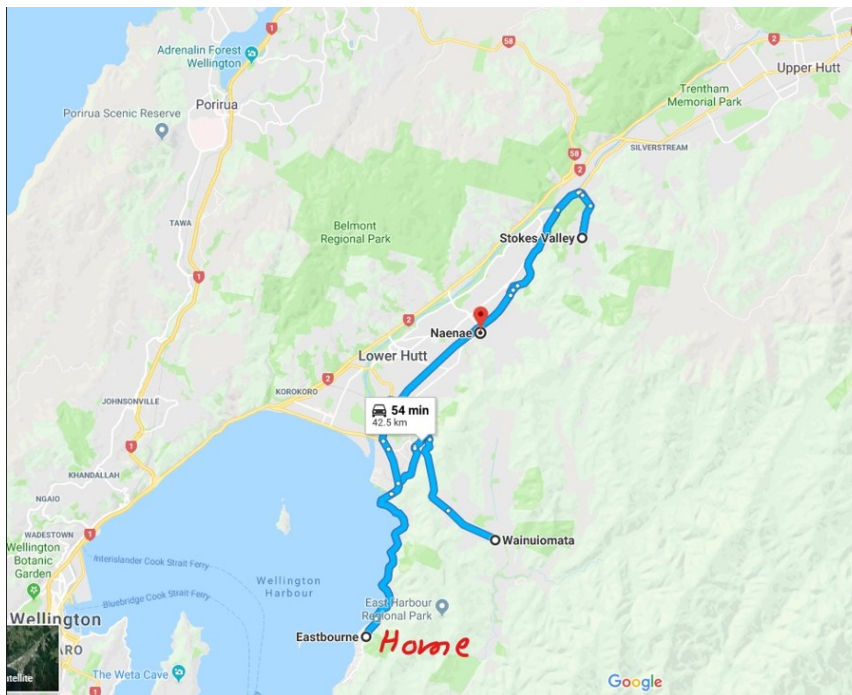
Location: /img_Narcos-1.001/vol_vol7/Users/Steve/Documents/Misc/dropoff.jpg

Eastbourne Library location on map

Modified: 2019-02-02 14:06:06 NZDT

Accessed: 2019-02-02 15:31:06 NZDT

Created: 2019-02-02 14:06:05 NZDT

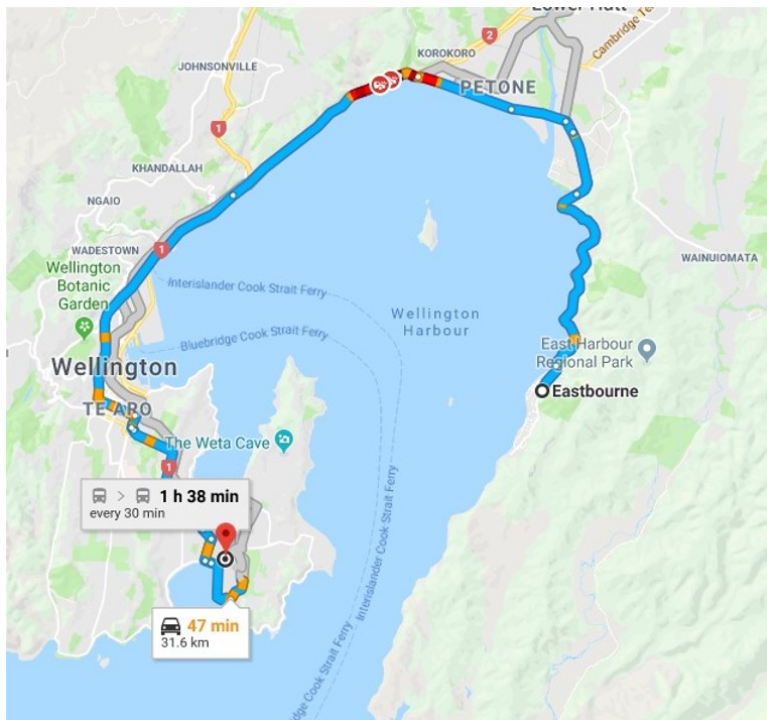


Location: /img_Narcos-1.001/vol_vol7/Users/Steve/Documents/Misc/Method run.jpg

Modified: 2019-01-31 10:22:24 NZDT

Accessed: 2019-01-31 16:04:13 NZDT

Created: 2019-01-31 10:21:24 NZDT



Location: /img_Narcos-1.001/vol_vol7/Users/Steve/Documents/Misc/airport crystals.jpg

Modified: 2019-01-31 10:25:18 NZDT

Accessed: 2019-01-31 16:04:13 NZDT

Created: 2019-01-31 10:25:18 NZDT



Image of Eastbourne Library

Location:
/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Local/Google/Chrome/User Data/Default/Cache/f_00007b



Location:

/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Local/Google/Chrome/User Data/Default/Cache/f_000033

Web Search

2019-01-28 20:34 – Viewed MetService (Wellington / Lower Hutt weather)
2019-01-28 20:55 – Searched “Discord” and downloaded DiscordSetup.exe
2019-01-28 22:59 – Searched “cutting agent” on Wikipedia
2019-01-28 23:00 – Searched “drug cutting”
2019-01-28 23:00 – Searched “cause effects”
2019-01-28 23:00 – Searched “cutting agent drug manufacturing”
2019-01-28 23:00 – Searched “what is meth cut with”
2019-01-29 01:04 – Searched “international drug routes”
2019-01-29 01:03 – Searched drug routes around Wellington
2019-01-31 15:55 – Searched “drug facts on crystal meth”
2019-02-02 14:01 – Searched “best places to trade drugs”

Task 6: Anti-Forensics

[List evidence relating to obfuscation, encryption, deletion, or data concealment.]

Discord Chat Logs

“Good Thinking, I already know how. Heard of steganography?”

“A way of hiding one image within another. There's a simple application called 'Image Steganography’”

“Ya.. I just told you about the tool :face_palm: Received it. Will check to see if it works and confirm soon.”

Location:

/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Roaming/Discord/Local Storage/leveldb/000006.txt

CCleaner

CCleaner Installer:

/img_Narcos-1.001/vol_vol7/Users/Steve/Downloads/ccsetup552.exe

Downloaded: 2019-01-30 12:40:51 NZDT

CCleaner Installed location: /img_Narcos-1.001/vol_vol7/Program
Files/CCleaner/CCleaner.exe

Last Accessed: 2019-02-02 15:38:42

Image Steganography

Installer: /img_Narcos-1.001/vol_vol7/Users/Steve/Downloads/Image
Steganography 1.5.2 Setup.exe

Downloaded: 2019-02-01 13:16:32 NZDT

Location:

/img_Narcos-1.001/vol_vol7/\Users\Steve\AppData\Roaming\CPascoe\Image
Steganography\ProgData\Image Steganography.exe

BNE.png



Metadata

Name:	/img_Narcos-1.001/vol_vol7/Users/Steve/Downloads/Misc/BNE.png
Type:	File System
MIME Type:	image/png
Size:	4376793
File Name Allocation:	Allocated
Metadata Allocation:	Allocated

Location: /img_Narcos-1.001/vol_vol7/Users/Steve/Downloads/Misc/BNE.png)

000000000006.bin

```
AB5X
AB5X
y|aHK
y|aHK
sucess
Success
|aHK
c 6[J
Stego
The location = C:\Users\Steve\Downloads\Misc \BNE
Pass equals Elchapo2
Success
```

BJ

Location:

/img_Narcos-1.001/vol_vol7/Users/Steve/AppData/Local/Packages/Microsoft.Off

ice.OneNote_8wekyb3d8bbwe/LocalState/AppData/Local/OneNote/16.0/cache/00000006.bin

Password equals Elchapo2 for the BNE.png image.

TrueCrypt encryption

Installer: /img_Narcos-1.001/vol_vol7/Users/Steve/Downloads/TrueCrypt Setup 7.1a.exe

Downloaded: 2019-01-30 12:27:43 NZDT

Encrypted file: /img_Narcos-1.001/vol_vol7/Users/Steve/Documents/secret

Windows user account deletion

(From volatility, run on DESKTOP-EN8JNOM-20190129-041409.dmp)

Registry data remnants from a deleted hive (user):

\\?\\C:\\Users\\JohnF\\ntuser.dat

Microsoft package paths: \\?\\C:

\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.WindowsStore_8wekyb3d8bbwe\\

```
C:\Users\Cyber\Downloads\John Primary\Day 1>python C:\Users\Cyber\Desktop\DigitalForensicTools\volatility-2.6.1\vol.py -f DESKTOP-EN8JNOM-20190129-041409.dmp --profile=Win10x64_17134 hivelist
Volatility Foundation Volatility Framework 2.6.1
Virtual Physical Name
0xfffff8b01151a500 0x0000000036933000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.Windows.ShellExperienceHost_cw5n1h2txyewy\\Settings\\settings.dat
0xfffff8b01151b700 0x0000000036939000 \\?\\C:\\ProgramData\\Microsoft\\Windows\\AppRepository\\Packages\\Microsoft.Windows.Cortana_1.0.7.17134_neutral_neutral_cw5n1h2txyewy\\ActivationStore.dat
0xfffff8b01152b500 0x0000000036b43000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.Windows.Cortana_cw5n1h2txyewy\\Settings\\settings.dat
0xfffff8b0115af500 0x0000000042121000 \\?\\C:\\ProgramData\\Microsoft\\Windows\\AppRepository\\Packages\\Microsoft.LockApp_10.0.17134.1_neutral_cw5n1h2txyewy\\ActivationStore.dat
0xfffff8b0115b8000 0x0000000054808000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.LockApp_cw5n1h2txyewy\\Settings\\settings.dat
0xfffff8b0115a7100 0x0000000010eaa8000 \\?\\C:\\ProgramData\\Microsoft\\Windows\\AppRepository\\Packages\\Microsoft.WindowsStore_11712.1001.23.0_x64_8wekyb3d8bbwe\\ActivationStore.dat
0xfffff8b011747c00 0x00000000033ab000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.WindowsStore_8wekyb3d8bbwe\\Settings\\settings.dat
0xfffff8b011252b00 0x00000000061d1000 \\?\\C:\\ProgramData\\Microsoft\\Windows\\AppRepository\\Packages\\Microsoft.SkypeApp_1213.274.0_x64_kzf8qxf38zg5c\\ActivationStore.dat
0xfffff8b01127a900 0x00000000628ba000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.SkypeApp_kzf8qxf38zg5c\\Settings\\settings.dat
0xfffff8b01179f000 0x0000000010e8c5000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.WindowsCommunicationsApps_8wekyb3d8bbwe\\Settings\\settings.dat
0xfffff8b01179bf00 0x000000007d8d1000 \\?\\C:\\ProgramData\\Microsoft\\Windows\\AppRepository\\Packages\\Microsoft.WindowsCommunicationsApps_17.8827.22055.0_x64_8wekyb3d8bbwe\\ActivationStore.dat
0xfffff8b01144c800 0x00000000106b50000 \\?\\C:\\ProgramData\\Microsoft\\Windows\\AppRepository\\Packages\\Windows.ImmersiveControlPanel_10.0.2.1000_neutral_neutral_cw5n1h2txyewy\\ActivationStore.dat
0xfffff8b01102b000 0x000000001f551000 \\?\\C:\\ProgramData\\Microsoft\\Windows\\AppRepository\\Packages\\Microsoft.Windows.Photos_2018.18011.15918.0_x64_8wekyb3d8bbwe\\ActivationStore.dat
0xfffff8b0118d5a00 0x0000000021000000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.Windows.Photos_8wekyb3d8bbwe\\Settings\\settings.dat
0xfffff8b0114bf600 0x00000000978d0000 \\?\\C:\\ProgramData\\Microsoft\\Windows\\AppRepository\\Packages\\Microsoft.Edge_42.17134.1.0_neutral_8wekyb3d8bbwe\\ActivationStore.dat
0xfffff8b0119a5300 0x00000000131c35000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.Edge_8wekyb3d8bbwe\\Settings\\settings.dat
0xfffff8b011203000 0x000000000e9d000 [no name]
0xfffff8b011203e00 0x00000000010a8000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.Edge_8wekyb3d8bbwe\\Settings\\settings.dat
0xfffff8b011205800 0x0000000009121000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.Edge_8wekyb3d8bbwe\\Settings\\settings.dat
0xfffff8b011356400 0x0000000011c84f000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.Edge_8wekyb3d8bbwe\\Settings\\settings.dat
0xfffff8b0112aab00 0x0000000011e16a000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.Edge_8wekyb3d8bbwe\\Settings\\settings.dat
0xfffff8b011363100 0x0000000011e3a9000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.Edge_8wekyb3d8bbwe\\Settings\\settings.dat
0xfffff8b011375d00 0x000000001262bc000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.Edge_8wekyb3d8bbwe\\Settings\\settings.dat
0xfffff8b011376f00 0x000000001246d2000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.Edge_8wekyb3d8bbwe\\Settings\\settings.dat
0xfffff8b01130e000 0x00000000120990000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.Edge_8wekyb3d8bbwe\\Settings\\settings.dat
0xfffff8b01139f600 0x00000000125168000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.Edge_8wekyb3d8bbwe\\Settings\\settings.dat
0xfffff8b01139e900 0x00000000128b4c000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.Edge_8wekyb3d8bbwe\\Settings\\settings.dat
0xfffff8b01141e000 0x000000001213e000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.Edge_8wekyb3d8bbwe\\Settings\\settings.dat
0xfffff8b011498b00 0x0000000029c0c000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.Edge_8wekyb3d8bbwe\\Settings\\settings.dat
0xfffff8b011498000 0x0000000029c40000 \\?\\C:\\Users\\JohnF\\AppData\\Local\\Packages\\Microsoft.Edge_8wekyb3d8bbwe\\Settings\\settings.dat
```

Command: python C:
\\Users\\Cyber\\Desktop\\DigitalForensicTools\\volatility-2.6.1\\vol.py -f
DESKTOP-EN8JNOM-20190129-041409.dmp --profile=Win10x64_17134 hivelist

VMWare tools

Autopsy shows that VMWare has been installed by its Virtual USB Hub Driver listed in external devices

VMWare root directory in: /img_Narcos-1.001/vol_vol7/Program Files/VMware/VMware Tools

External Disk Drives for Data Storage

Autopsy analysis identified two external disk drives connected to Desktop:

Seagate RSS LLC Backup Plus Slim Portable Drive 1 TB 2019-01-31 16:04:08

Western Digital Technologies, Inc. Elements Portable (WDBUZZG) 2019-02-01 15:41:46

Web Search

2019-02-01 – Searched “image steganography download”

2019-01-30 – Searched “TrueCrypt”

Web history

2019-01-30 12:27:43 - True crypt setup 9.1.exe

2019-01-31 15:56:56 - ccsetup552.exe

2019-02-01 13:16:32 - image steganography 1.5.2 setup.exe

Findings Interpretation

[For each examination task, state what the team believes the catalogued evidence shows. This is the team's collective expert interpretation. Every claim must be grounded in specific catalogued evidence. Acknowledge limitations and identify alternative explanations where they exist. Use confidence language: 'established', 'suggested', or 'cannot be determined'.]

[Every team member must be able to defend any conclusion in this section during the interview. Joint authorship means joint accountability.]

Task 1: User Attribution

[State the team's interpretation of what the catalogued evidence for this task shows. Indicate confidence (established / suggested / cannot be determined). Acknowledge limitations and alternative explanations. Reference the catalogue items that support each claim.]

The desktop computer (EV-001) is assumed to be operated by a primary user "Steve." This is backed by multiple sources:

Analysis of the OneNote cache file 00000006.bin identified references suggesting that Steve's surname may be "Kowhai," indicating the desktop user's full name could potentially be identified as Steve Kowhai.

User profile directory at C:\Users\Steve\ contains Desktop, Documents, Downloads, Pictures, and AppData subdirectories, consistent with an active user account.

SAM registry analysis shows a user account "Steve" with SID S-1-5-21-1474204758-2504895174-1356074821-1001, created on 2019-01-28 19:35:06 UTC, with 18 recorded logins. The last login occurred on 2019-02-02 02:38:22 UTC.

NTUSER.DAT registry hive contains user-specific settings, confirming Steve as the primary account holder. The NTUSER.DAT registry in Steve's home directory reveals the processes that are being run using UserAssist (Administrative privileges). We can see that he is infact running CCleaner, Image Steganography, TrueCrypt, and has Discord installed. This confirms Steve's intent to communicate with a way to potentially hide the content of his messaging, while attempting to cover his tracks afterwards using CCleaner.

Volatility memory scan on desktop to show command line processes reveals that the desktop owner: Steve, was using Discord, VMTools, and Baidu anti-virus which indicate communication software, virtualisation tools and anti-virus software were installed and running. Discord was channel of communication, and the anti-virus suggest attempt to secure himself from potential nefarious actors including websites visited,

Extracted images from Autopsy show various drug-related paraphernalia such as methamphetamine, crystals, cash, syringe, lighter, teaspoon, zip-lock bags. Further investigation reveals an image of mongrel a mob jacket, indicating some relevance and potential gang association or affiliation.

Desktop wallpaper at TranscodedWallpaper shows a crayfish image. The same image appears in Steve's Pictures folder, indicating personal customisation. Being used as a wallpaper indicating interest in crayfish.

Proton Mail account shows the username crayfish1980, which matches the crayfish theme of the desktop wallpaper, further linking Steve to this online identity.

Limitations: Physical address, or government identification was recovered. The user "Steve" could be an alias rather than the individual's legal name. No further specific information on Steve as an individual i.e. a photo representation.

Confidence: Suggested.

Task 2: Relationship with John Fredricksen

[State the team's interpretation of what the catalogued evidence for this task shows. Indicate confidence (established / suggested / cannot be determined). Acknowledge limitations and alternative explanations. Reference the catalogue items that support each claim.]

John Fredricksen is connected to Steve (the desktop user) through documented and logged communications. This is supported by the evidence:

Discord chat logs contain direct conversation between Steve and John Fredricksen. The chat includes:

- Reference to "10 keys" (slang for kilograms of drugs) to be delivered to Wellington.
- Discussion of steganography for concealing information, which shows intent of anti-forensics.
- Meeting instructions: "Meet at the Eastbourne library and if we miss each other come to 666 Rewera Avenue, Petone".

Proton Mail email exchange shows Steve accessed Proton Mail. This reveals intent to hide their communications. Also found (in Task 3) a Proton Mail named "heresjonny1@protonmail.com" indicating potentially John's Proton Mail indicating use of this as a communication channel and hence Steve and John being associated.

Both Steve (desktop) and John Fredricksen (laptop memory dump from Day 2) had Discord installed and running.

Both had TrueCrypt installed for encryption.

Both had VMware Tools installed, indicating both systems are virtual machines

Limitations: The relationship between Steve and John Fredricksen is based on inferred associations from Discord communications and system artifacts. The full identity of Discord users cannot be conclusively verified, and usernames may not directly represent real-world individuals. The full name "John Fredricksen" does not appear in the desktop evidence. The connection is inferred from the laptop naming convention and the context of the case. TrueCrypt, Proton Mail, and VMware Tools are legitimate applications and their presence does not independently establish criminal activity.

Confidence: Suggested.

Task 3: Drug Trafficking Network

[State the team's interpretation of what the catalogued evidence for this task shows. Indicate confidence (established / suggested / cannot be determined). Acknowledge limitations and alternative explanations. Reference the catalogue items that support each claim.]

Steve and John Fredricksen were participants in a drug trafficking network. This is established by the following evidence:

Discord communications:

"New supplier eh? Definitely Interested! Can I get 10 keys of it delivered to Wellington" - "10 keys" is a known slang term for 10 kilograms of drugs (typically methamphetamine).

"It contains some information regarding how I see this working out" indicates an operational plan for drug distribution.

"Meet at the Eastbourne library and if we miss each other come to 666 Rewera Avenue, Petone" matches Jane Esteban's intercepted statements about delivery locations.

This discord chat log was inside of the Desktop under the user Steve and it was confirmed that this conversation was happening with John (Steve's associate) as both the desktop user (Steve) and John's laptop have Discord.exe processes that access the specific channel or (chat room) bring "539550615072800768" which is the matching channel ID shown in the evidence catalog.

Discord application artifact shows that discord being on Steve's desktop and also John's Laptop further indicating the presence of discord being a communication tool. Further support of the claim above.

Day 2 analysis of John's laptop memory identified active instances of Discord, TrueCrypt, and VM tools. The presence of these applications suggests potential use of encrypted storage and communication methods. Discord activity indicates that communication may have been occurring at the time the memory image was captured.

Day 3 analysis of John's laptop again identified Discord processes running in memory, supporting the assessment that Discord was used as an ongoing communication platform.

Day 4 analysis identified Discord activity alongside image steganography-related tools/processes. This may indicate the potential use of steganographic techniques to conceal data within image files transmitted through Discord communications with both Steve and John. Image steganography was on both Steve and John's laptop.

Analysis also identified the presence of TrueCrypt artifacts on both John's laptop and Steve's desktop. The shared presence of encryption software may suggest the use of encrypted containers or files for operational security purposes via communication channels.

Evidence of access to the encrypted email service Proton Mail, including the email accounts "crayfish1980@protonmail.com" and "heresjohnny1@protonmail.com" was identified on Steve's system. The use of an encrypted email provider is consistent with attempts to enhance privacy and operational security through end-to-end encrypted communications. In this case, it's most likely that that crayfish1980@protonmail.com indicates Steve's proton mail as he has wallpaper was a crayfish and "heresjohnny1@protonmail.com" indicates John's proton mail as it contains the name "johnny".

Web search activity indicates repeated queries related to drug terminology, manufacturing processes, and trafficking routes, including references to methamphetamine, cutting agents, and international drug routes. This suggests involvement in researching drug distribution and supply chain methods, supporting the assessment of a drug trafficking network.

A web history artifact identified that a file named "flightbookings.PNG" was stored within the user's Documents directory. The file originated from a Discord CDN URL (cdn.discordapp.com), indicating that it was likely received or accessed through Discord communications. The flight-related nature of the image may suggest coordination of travel arrangements, potentially relating to movement between Brisbane and Wellington in connection with the investigation.

Limitations: The findings are based on inferred associations from digital artifacts rather than direct identification of individuals. Usernames and accounts cannot be definitively linked to real-world identities without external verification. Some terms and search queries may have alternative meanings depending on context, and application usage (encryption tools) does not on its own confirm illegal activity.

Confidence: Suggested

Task 4: Customers

[State the team's interpretation of what the catalogued evidence for this task shows. Indicate confidence (established / suggested / cannot be determined). Acknowledge limitations and alternative explanations. Reference the catalogue items that support each claim.]

Various photos were recovered during exploration of EV-001. Many of these photos contain people's faces and people partaking in social activities. These photos could include John, Steve, Jane, or people relating to the individuals involved. But this provides a suggested confidence.

Negative findings: No customer names, phone numbers, addresses, transaction records, payment confirmations (bank transfers, cryptocurrency), or buyer communications were identified in:

- Discord chat logs
- Browser history and cache
- Windows Mail app
- Downloaded documents
- Registry artifacts
- Recycle Bin contents

An image of a woman was identified. The image appears to show a possible "before and after" comparison; however, the identity of the woman and the context of the image could not be confirmed.

Another image showed four men in a pool, with three faces blurred. The blurring may indicate an attempt to hide identities or associations with individuals related to the investigation.

A further image showed a man standing next to another person whose face was blurred, possibly to conceal identity. However, the identities and context could not be confirmed.

A file called client.ods exists in the memory of Steve's desktop and John's laptop, however we were unable to recover the contents of this file, which potentially contains information on customers of John and Steve.

Limitations: No definitive evidence of customers, transactions, or exchanges involving Steve or John was identified across the recovered datasets.

Confidence: Cannot be determined

Task 5: Timeline and Future Plans

[State the team's interpretation of what the catalogued evidence for this task shows. Indicate confidence (established / suggested / cannot be determined). Acknowledge limitations and alternative explanations. Reference the catalogue items that support each claim.]

Phase 1: Account Creation and Initial Communication (January 28, 2019)

The operation began on **January 28, 2019**, when the desktop user account "Steve" was created at **19:35:06 NZDT** (SAM registry). Within hours, at **20:18-20:21**, John Fredricksen's laptop executed DiscordSetup.exe, followed by the first run of Discord.exe at **20:21:15** with the "--squirrel-first-run" flag. This indicates that Discord was chosen as the primary communication channel between Steve and John Fredricksen. Discord logs would later contain direct evidence of their trafficking agreement.

At **20:34**, Steve viewed MetService weather for Wellington/Lower Hutt (browser history). This seemingly innocuous activity takes on significance when viewed in context: weather monitoring is a common precursor to drug importation operations, as maritime or air

transport can be affected by conditions. This suggests Steve was already considering logistical factors for an impending shipment.

By **20:55**, Steve searched for and downloaded Discord (browser history), supporting his intent to establish real-time communication with John Fredricksen.

Phase 2: Research and Planning (January 28-29, 2019)

Between **22:59 and 23:00** on January 28, Steve conducted a concentrated series of searches (browser history):

- "cutting_agent" (Wikipedia)
- "drug cutting"
- "cutting agent drug manufacturing"
- "what is meth cut with"

These searches are indicative of drug trafficking intent. *Cutting agents* are substances used to dilute pure *methamphetamine* before resale, increasing volume and profit margin. The fact that Steve researched this topic immediately after establishing communication with John Fredricksen suggests he was planning the downstream distribution of the anticipated shipment.

On **January 29, between 01:03 and 01:04**, Steve's searches shifted to logistics: "drug routes around Wellington" and "international drug routes" (browser history). The timing of late night/early morning is consistent with covert planning activity. These searches directly preceded the flight booking that would later appear, indicating Steve was mapping potential importation pathways into Wellington.

Phase 3: Anti-Forensics Tool Acquisition (January 30-31, 2019)

Recognising the need to conceal their activities, Steve began acquiring anti-forensic tools. On **January 30 at 12:27**, he downloaded TrueCrypt Setup 7.1a.exe (browser history). TrueCrypt provides full-disk and container encryption, capable of making evidence inaccessible without a passphrase. An encrypted file was found at C:\Users\Steve\Documents\secret, confirming its use.

At **12:40** on the same day, Steve downloaded ccsetup552.exe (CCleaner) (browser history). CCleaner can securely delete browsing history, temporary files, and registry entries, essentially destroying forensic evidence of his searches and communications. The fact that Steve acquired both encryption and wiping tools within 13 minutes of each other demonstrates deliberate, methodical anti-forensic activities.

On **January 31 at 16:04:08**, Steve connected an external Seagate 1TB drive to the desktop (external devices). External storage is commonly used to isolate evidence from the primary system, allowing it to be physically removed or destroyed if necessary.

Phase 4: Operational Preparation (January 31, 2019)

Between **10:21 and 10:25** on **January 31**, Steve created two critical files:

- Method run.jpg
- airport crystals.jpg

The Method run.jpg file shows a driving route (54 minutes), passing through Eastbourne, Naenae, Stokes Valley, and Wainuiomata. The word "home" is also present near Eastbourne, suggesting it may be a reference point in the route.

The airport crystals.jpg file shows a driving route from the wellington airport to Eastbourne showing the planning and preparation that was taken and that Eastbourne being a meet up stop for Steve and John from which is supported with the Discord chat logs.

At **15:55**, Steve searched for "drug facts on crystal meth" (browser history). This may represent last-minute research before finalising operational plans, or an attempt to understand potential legal penalties.

Phase 5: Steganography and Final Planning (February 1-2, 2019)

On **February 1 at 13:16**, Steve downloaded Image Steganography 1.5.2 Setup.exe (browser history). Steganography allows hidden data to be embedded within image files, evading signature-based detection. Discord chat logs confirm the purpose: "A way of hiding one image within another". Steve later received BNE.png via ProtonMail - an image likely containing hidden data, as suggested by its large file size.

At **15:41:46** on February 1, Steve connected a second external drive (Western Digital) (external devices). The connection of two different external drives suggests data segregation - possibly one for encrypted communications and another for operational documents.

On **February 2**, Steve executed the final planning phase:

- **14:01** - Searched "best places to trade drugs" (browser history). This indicates Steve was actively seeking distribution channels for the incoming shipment.
- **14:06** - Created dropoff.jpg, a map showing Eastbourne Library. This directly matches the Discord meeting instruction: "Meet at the Eastbourne library and if we miss each other come to 666 Rewera Avenue, Petone". The creation of a visual map suggests operational planning for the physical handoff.
- **15:28** - Created flightbookings.PNG, showing Brisbane to Wellington return flights. This file was received via Discord CDN (originally named Steve_K.PNG), indicating John Fredricksen sent it to Steve. The flight bookings corroborate John Fredricksen's travel from Brisbane to Wellington for the importation.
- **15:38:42** - Steve last accessed CCleaner, potentially attempting to wipe evidence after completing his planning.

Phase 6: Capture and Aftermath

On **January 30, 2019 at 02:03:49 UTC** (approximately 15:03 NZDT), a memory dump was captured from John Fredricksen's laptop at Wellington Airport - consistent with his interception by Customs. The dump shows Discord.exe actively running, confirming that communications were ongoing at the moment of capture.

Registry remnants of a deleted user "JohnF" appear in the memory dump, indicating attempts to remove John Fredricksen's user account from the desktop after his arrest - consistent with post-capture destruction of evidence.

The last login for user "Steve" occurred on **February 2, 2019 at 02:38:22 UTC** (SAM registry), approximately two days before the flight booking date. This suggests Steve ceased activity on the desktop after finalising plans, possibly in anticipation of the importation or in response to news of John Fredricksen's interception.

Limitations: No information was recovered from before January 28, initial contact between characters John and Steve is unknown, planning or communications occurring before the desktop user account was created cannot be recovered, the full duration of the trafficking operation (previous importation attempts) remain unknown.

Confidence: Established

Task 6: Anti-Forensics

[State the team's interpretation of what the catalogued evidence for this task shows. Indicate confidence (established / suggested / cannot be determined). Acknowledge limitations and alternative explanations. Reference the catalogue items that support each claim.]

Steve and John Fredricksen actively employed multiple anti-forensic techniques to conceal their activities. This is supported by the evidence:

TrueCrypt:

TrueCrypt provides full-disk or container encryption, making evidence inaccessible without the passphrase.

Encryption (TrueCrypt):

Steve downloaded TrueCrypt Setup 7.1a.exe on 2019-01-30 12:27:43 NZDT

An encrypted file exists at C:\Users\Steve\Documents\secret

TrueCrypt was also running on John Fredricksen's laptop (from volatility memory dump), this implies that they both (Steve and John) needed it to decrypt valuable information that would be communicated between the two characters.

Steganography:

Discord chat explicitly discusses steganography: *"Good Thinking, I already know how. Heard of steganography?"*

"A way of hiding one image within another", "A way of hiding one image within another. There's a simple application called 'Image Steganography'".

This indicated the intent of hiding information using steganography and as already stated, image steganography tool was present in Steve desktop and John's laptop hence further supports this claim.

Steve downloaded Image Steganography 1.5.2 Setup.exe on 2019-02-01 13:16:32 NZDT

In the OneNote cache file 000000000006.bin, a reference was identified to the location C:\Users\Steve\Downloads\Misc\BNE, indicating the relevance of the BNE image. The cache also contains the text "Pass equals Elchapo2", suggesting a possible password associated with the file. Additionally, the keyword "stego" appears within the cache data, indicating potential use of steganography. These artifacts show that the BNE image has been likely used for hidden data storage, consistent with possible attempts to conceal information.

Image BNE.png was sent via ProtonMail and likely contains hidden data.

Steganography conceals data within image files, evading signature-based detection.

CCleaner:

Steve downloaded ccsetup552.exe on 2019-01-30 12:40:51 NZDT.

CCleaner was installed at C:\Program Files\CCleaner\CCleaner.exe with last access on 2019-02-02 15:38:42. CCleaner can securely delete browsing history, temporary files, and registry entries, destroying forensic evidence. This shows that a file-wiping tool was installed and used on the system, indicating potential attempts to remove or conceal digital evidence by clearing system and browsing artifacts.

User Account Deletion:

Registry remnants of a deleted user "JohnF" appear in the memory dump:

```
\\?\C:\Users\JohnF\ntuser.dat  
\\?\C:\Users\JohnF\AppData\Local\Packages\Microsoft.WindowsStore_8wekyb3d8bbwe\
```

Attempts were made to remove John Fredricksen's user account from the desktop, consistent with post-arrest destruction of evidence as the presence of these remnants within memory mean that a Windows user accounted with JohnF previously existed on the desktop system.

Virtual Machine Environment:

Multiple VMware Tools processes running on the desktop: vmacthlp.exe, vmtoolsd.exe (two instances), VGAAuthService.exe

VMware root directory present at C:\Program Files\VMware\VMware Tools

Virtual USB Hub Driver listed in external devices

John Fredricksen's laptop also showed VMware Tools running

Running the operating system as a virtual machine allows the user to:

- Isolate illegal activities from the host OS
- Destroy the VM file to instantly erase all evidence
- Create snapshots before risky operations

The physical host machine was not seized, potentially leaving key evidence undiscovered.

External Storage Devices

Two external drives were connected to the desktop:

Seagate Backup Plus Slim 1 TB (connected 2019-01-31)

Western Digital Elements Portable (connected 2019-02-01)

Evidence may have been stored on or transferred to external media that were not seized.

Browser Search for Anti-Forensic Tools

Search for "steganography download" on 2019-02-01

Search for "TrueCrypt" on 2019-01-30

Demonstrates deliberate research into anti-forensic methods.

Limitations: The analysis is based only on data recovered from disk images and memory dumps within the time of the investigation. Some evidence may be incomplete or missing because it could not be recovered. Findings are based only on available artifacts, and other activity may have occurred that was not captured. Some deleted or encrypted data may also be missing, so additional activity may not be visible.

Confidence: Established

Date: 22/05/2026